

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11
- 12
- 13
- 14
- 15
- 16
- 17
- 18
- 19
- 20
- 21
- 22
- 23

Metabolic coupling between soil aerobic methanotrophs and denitrifiers in rice paddy fields

Kang-Hua Chen^{1,2}, Jiao Feng^{1,2*}, Paul L. E. Bodelier³, Ziming Yang⁴, Qiaoyun Huang^{1,2},
Manuel Delgado-Baquerizo⁵, Peng Cai^{1,2}, Wenfeng Tan², Yu-Rong Liu^{1,2*}

¹National Key Laboratory of Agricultural Microbiology and College of Resources and Environment, Huazhong Agricultural University, Wuhan, 430070, China

²State Environmental Protection Key Laboratory of Soil Health and Green Remediation and Hubei Key Laboratory of Soil Environment and Pollution Remediation, Huazhong Agricultural University, Wuhan, 430070, China

³Department of Microbial Ecology, Netherlands Institute of Ecology (NIOO-KNAW), PO Box 50, 6700 AB, Wageningen, The Netherlands

⁴Department of Chemistry, Oakland University, Rochester, MI, 48309, USA

⁵Laboratorio de Biodiversidad y Funcionamiento Ecosistémico, Instituto de Recursos Naturales y Agrobiología de Sevilla (IRNAS), CSIC, Sevilla, 41012, Spain

***Corresponding authors**

Jiao Feng; E-mail: fengjiao@mail.hzau.edu.cn;

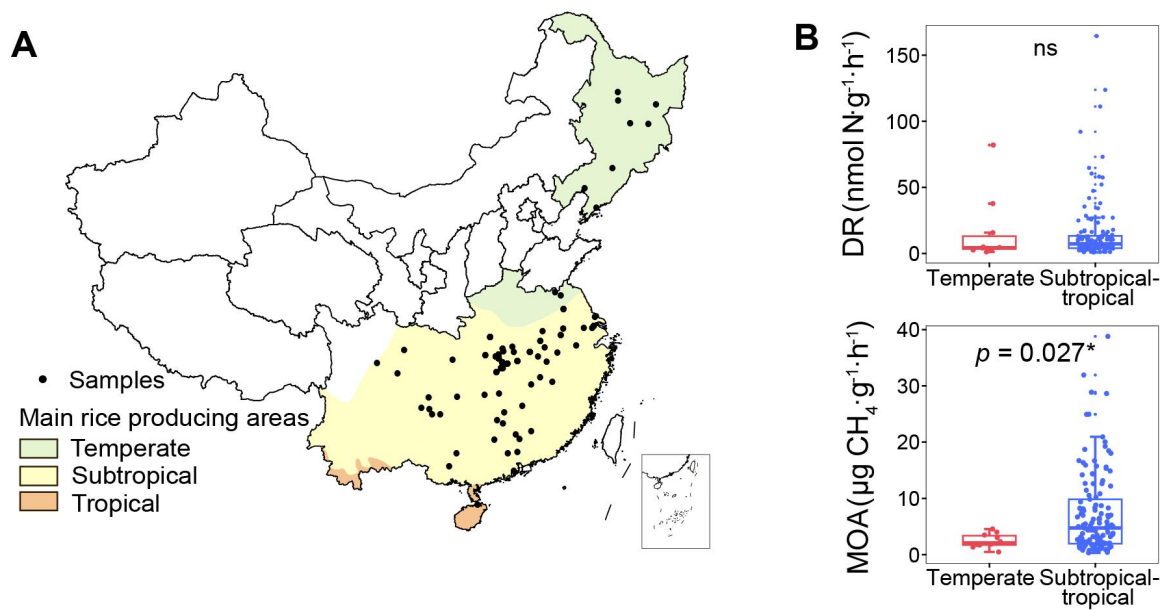
Yu-Rong Liu; E-mail: yrliu@mail.hzau.edu.cn

This PDF file includes:

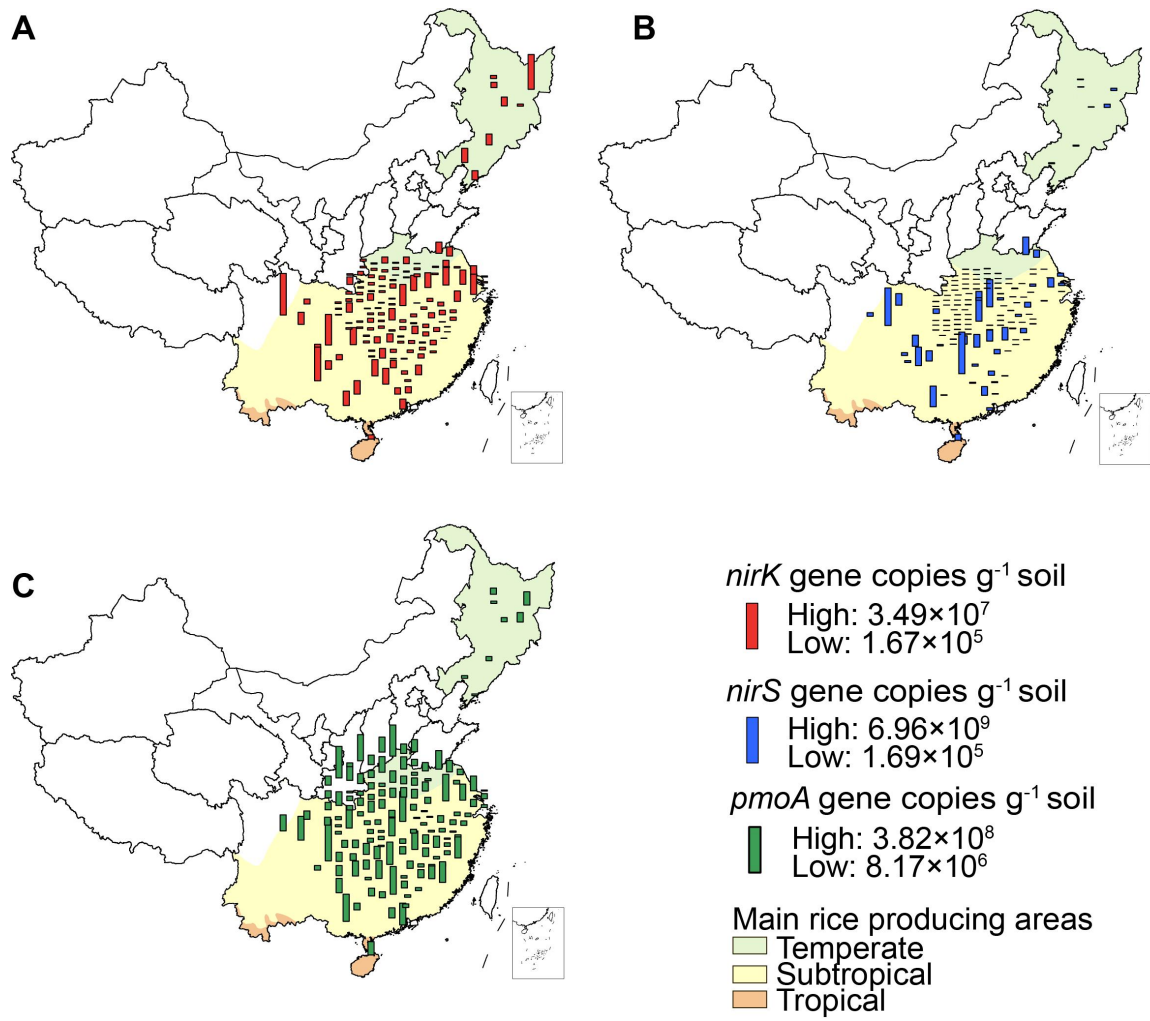
Supplementary Figs. 1 to 20

Supplementary Tables 1 to 7

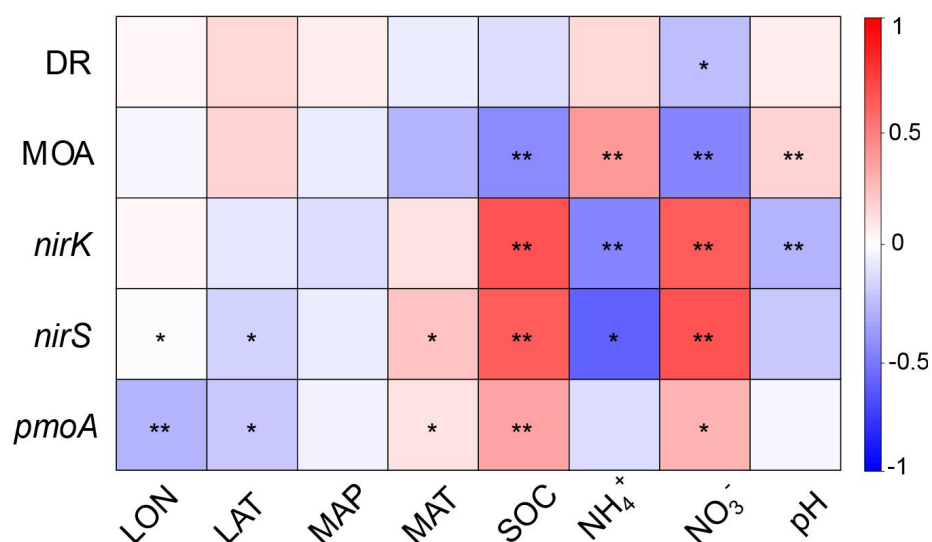
Supplementary References



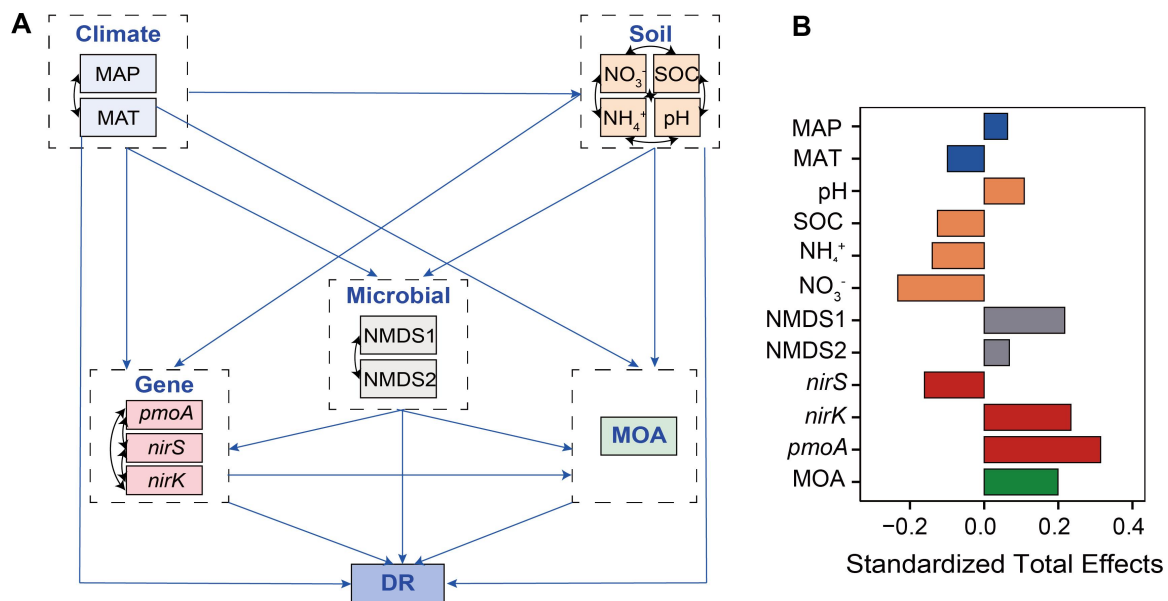
Supplementary Fig. 1 The distribution of sampling sites and the activities of denitrification (DR) and methane oxidation (MOA) across main rice-producing areas of China. **A** A total of 139 representative soil samples were collected, spanning a > 3300 km transect from the northeast to the south of China. The transect encompasses diverse climatic zones, including temperate, subtropical and tropical regions; **B** The differences of DR and MOA between temperate and subtropical-tropical regions. The box plots display the interquartile range, comprising the first quartile, median, and third quartile, while the whiskers extend from the minimum to the maximum values. * indicates statistically significant differences based on the two-sided T-Test of $p < 0.05$. Source data are provided as a Source Data file.



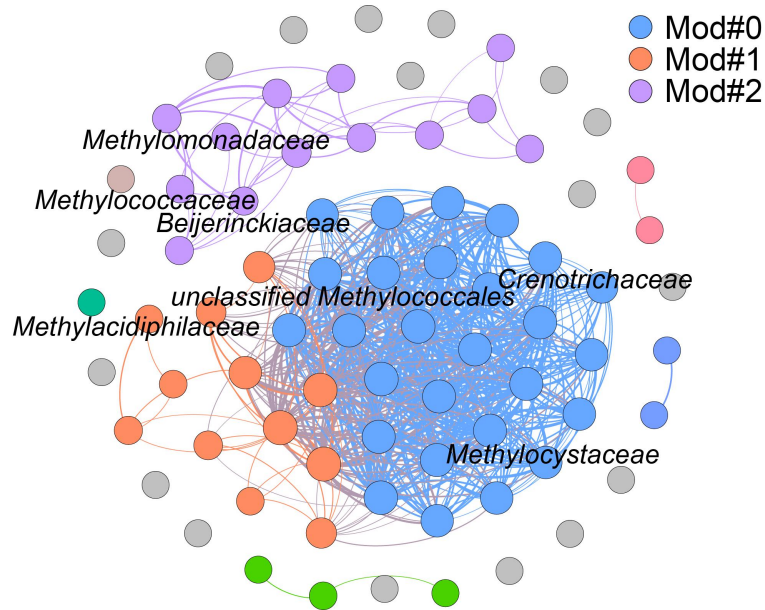
Supplementary Fig. 2 The abundance of genes associated with denitrification (*nirK* and *nirS*) and methane oxidation (*pmoA*) across climate zones of main rice-producing areas of China. A-C The distribution patterns of *nirK*, *nirS* and *pmoA* genes.



Supplementary Fig. 3 Pearson correlations of activities and genes associated with methane (CH₄) oxidation and denitrification with environmental factors. DR, denitrification rate; MOA, CH₄-oxidizing activity; LON, longitude; LAT, latitude; MAP, mean annual precipitation; MAT, mean annual temperature; SOC, soil organic carbon. * indicated $p < 0.05$ and ** indicated $p < 0.01$. Exact p -values and Source data are provided as a Source Data file.

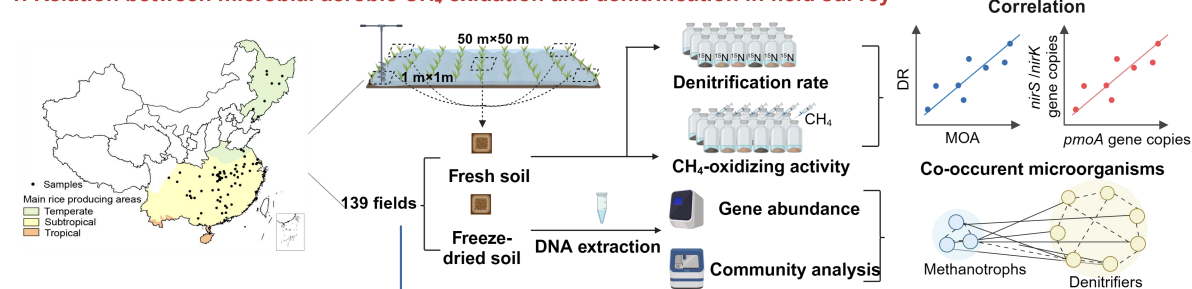


Supplementary Fig. 4 The rationale for the associations of environmental factors and their influences on soil denitrification rate (DR). **A** A priori generic structural equation model used in this study; **B** The standardized total effects of different environmental variables on DR. The model evaluated the effects of climatic (MAP and MAT), soil properties (SOC, NH₄⁺, NO₃⁻ and soil pH), methane-oxidizing activity (MOA) and microbial attributes (NMDS1, NMDS2 and functional gene abundances) on DR. MAP, mean annual precipitation; MAT, mean annual temperature; SOC, soil organic carbon; NMDS1 and NMDS2 represents the two axes of a nonmetric multidimensional scaling (NMDS) analysis. Source data are provided as a Source Data file.

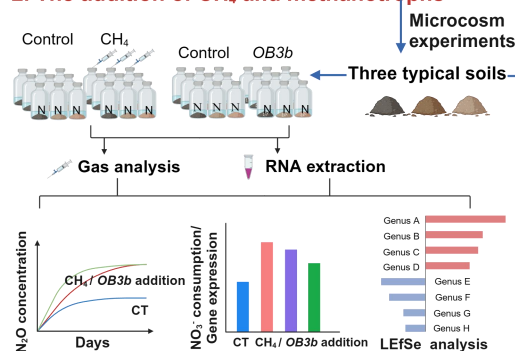


Supplementary Fig. 5 Methanotrophs associated with the denitrification process.
 Network analyses showing the co-occurrence patterns among methanotrophs and denitrifiers.
 A total of 71 denitrifying and 7 methanotrophic families co-occurred significantly across the
 main rice-producing areas of China. Source data are provided as a Source Data file.

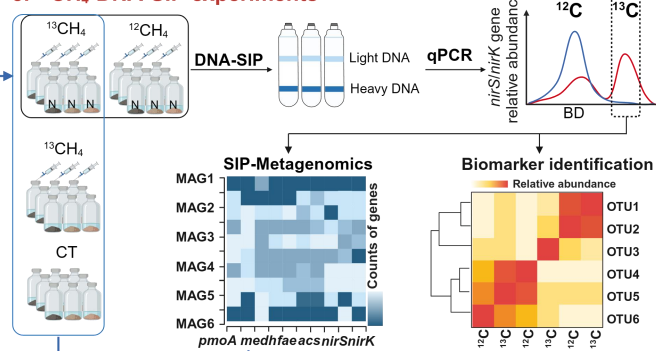
1. Relation between microbial aerobic CH₄ oxidation and denitrification in field survey



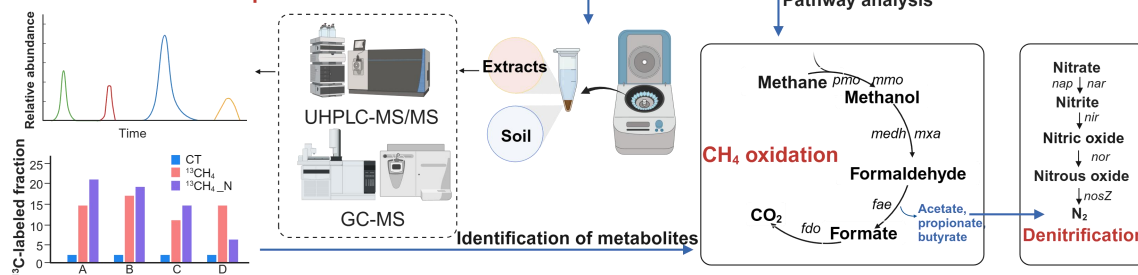
2. The addition of CH₄ and methanotrophs



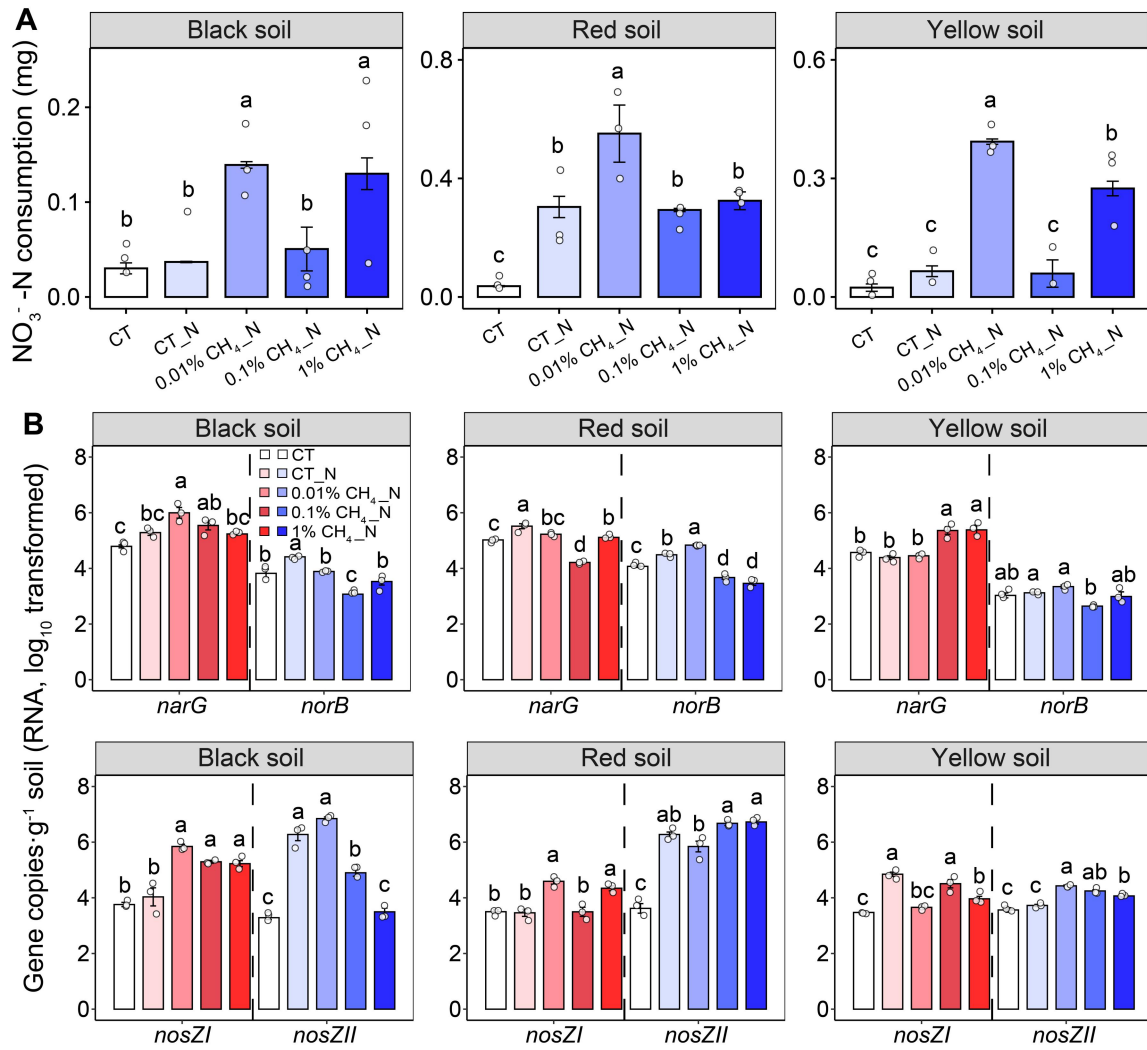
3. ¹³CH₄-DNA-SIP experiments



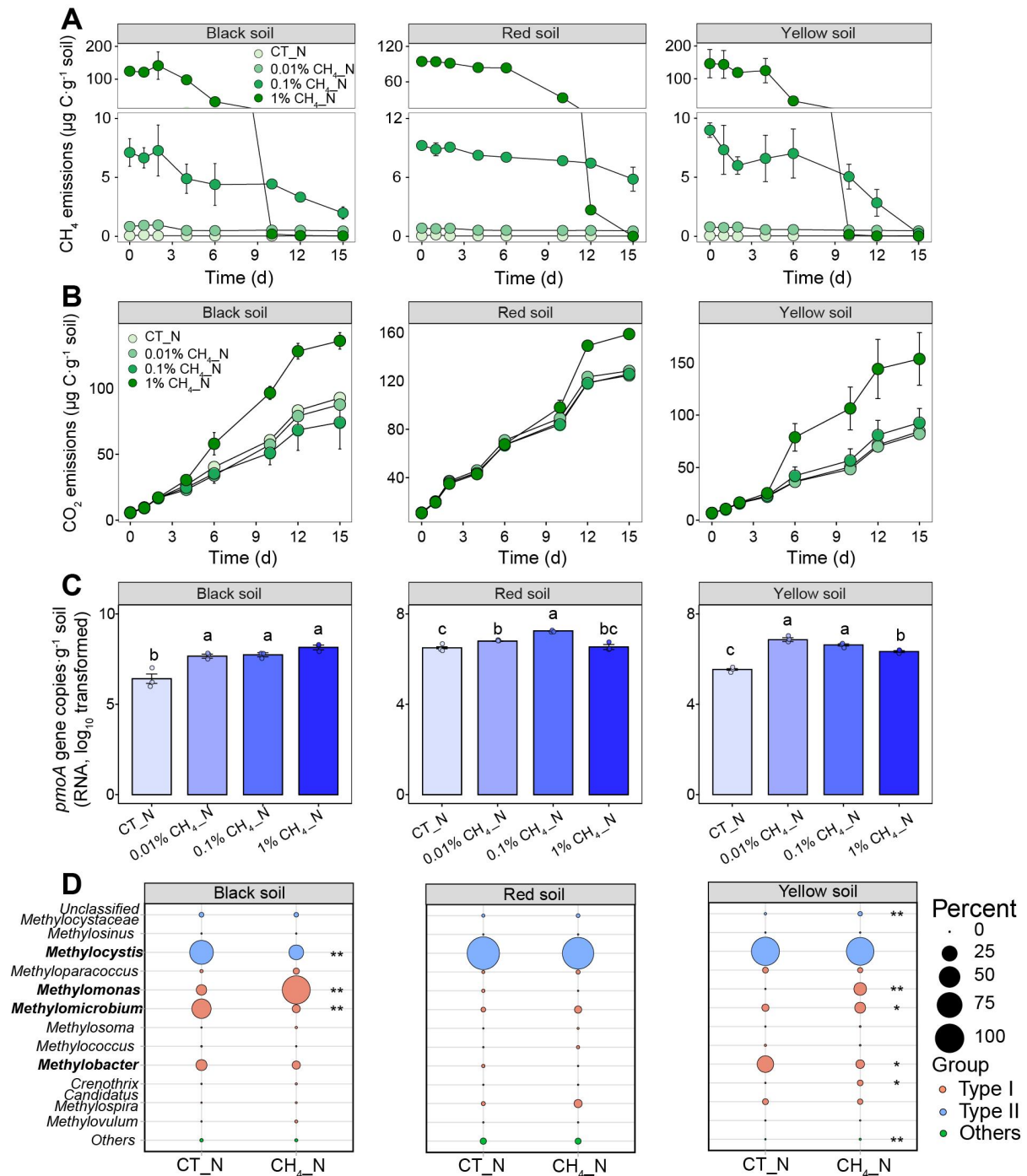
4. ¹³C-metabolomics experiments



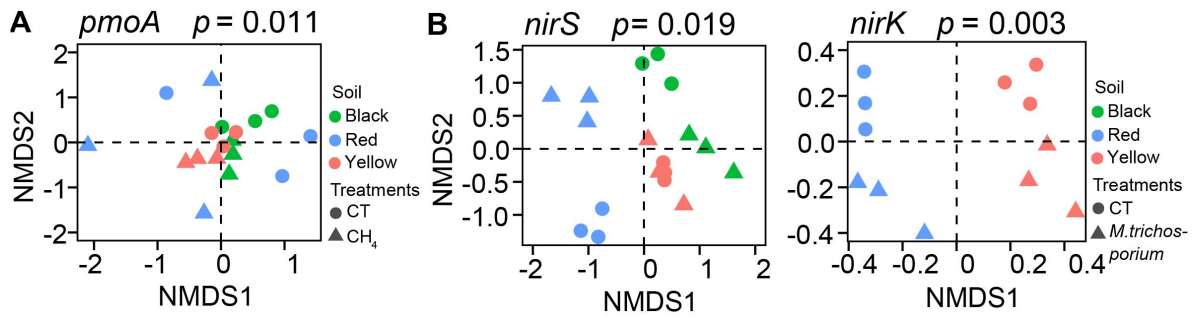
Supplementary Fig. 6 Workflow of the field survey and microcosm experiments in rice paddy fields.



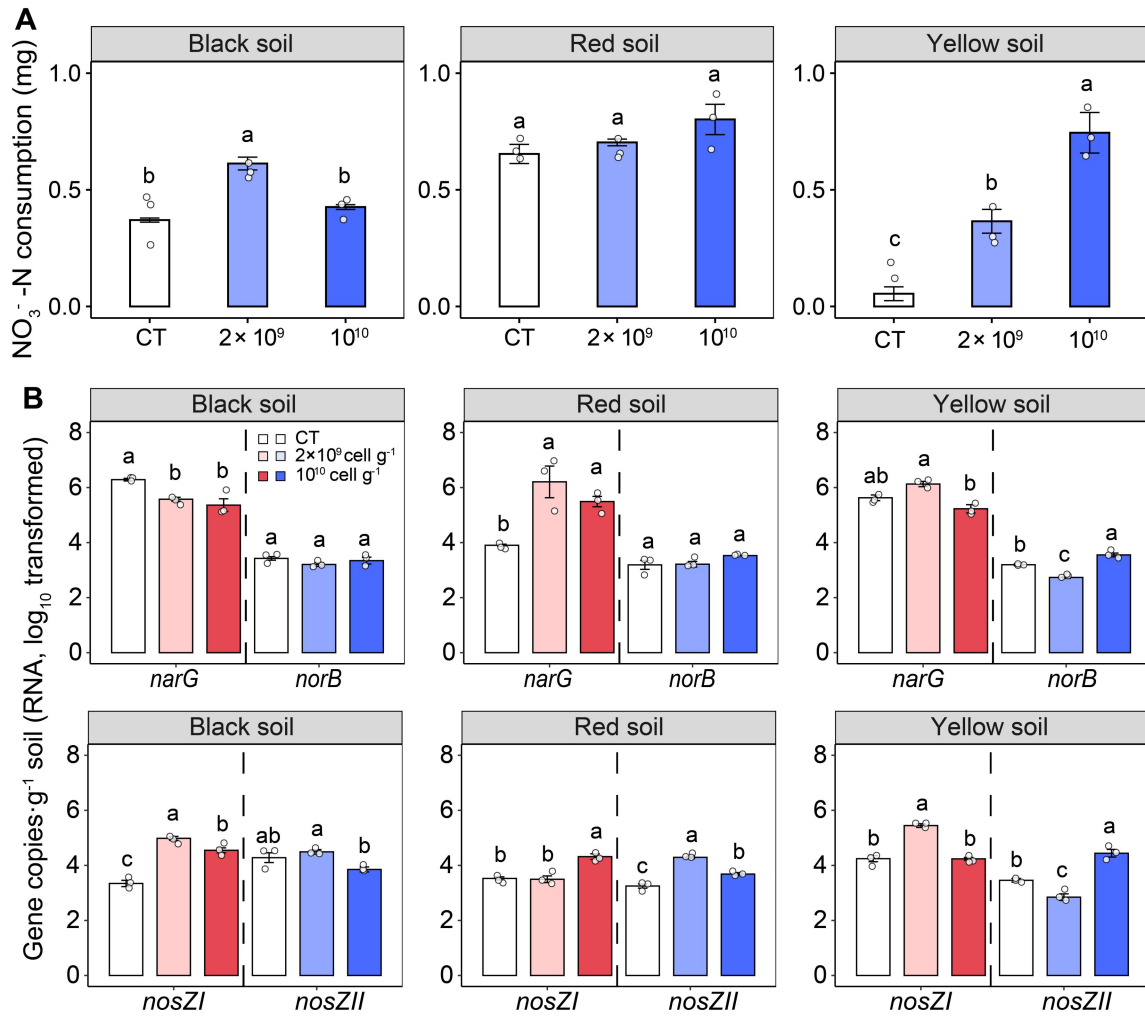
Supplementary Fig. 7 Effects of methane (CH_4) addition on microbial denitrification of three typical paddy soils. **A** Variations in consumption of nitrate (NO_3^- -N); and **B** Changes in denitrification gene (*narG*, *norB*, *nosZI* and *nosZII*) expression using RNA reverse transcription. The error bar represents the standard error of triplicate samples, and data are presented as mean values $\pm$ standard error. Different lowercase letters indicate significant differences between the soils with different CH_4 concentrations ($p < 0.05$; $n = 3$; one-way ANOVA followed by two-sided Tukey post hoc test). Exact p -values and Source data are provided as a Source Data file.



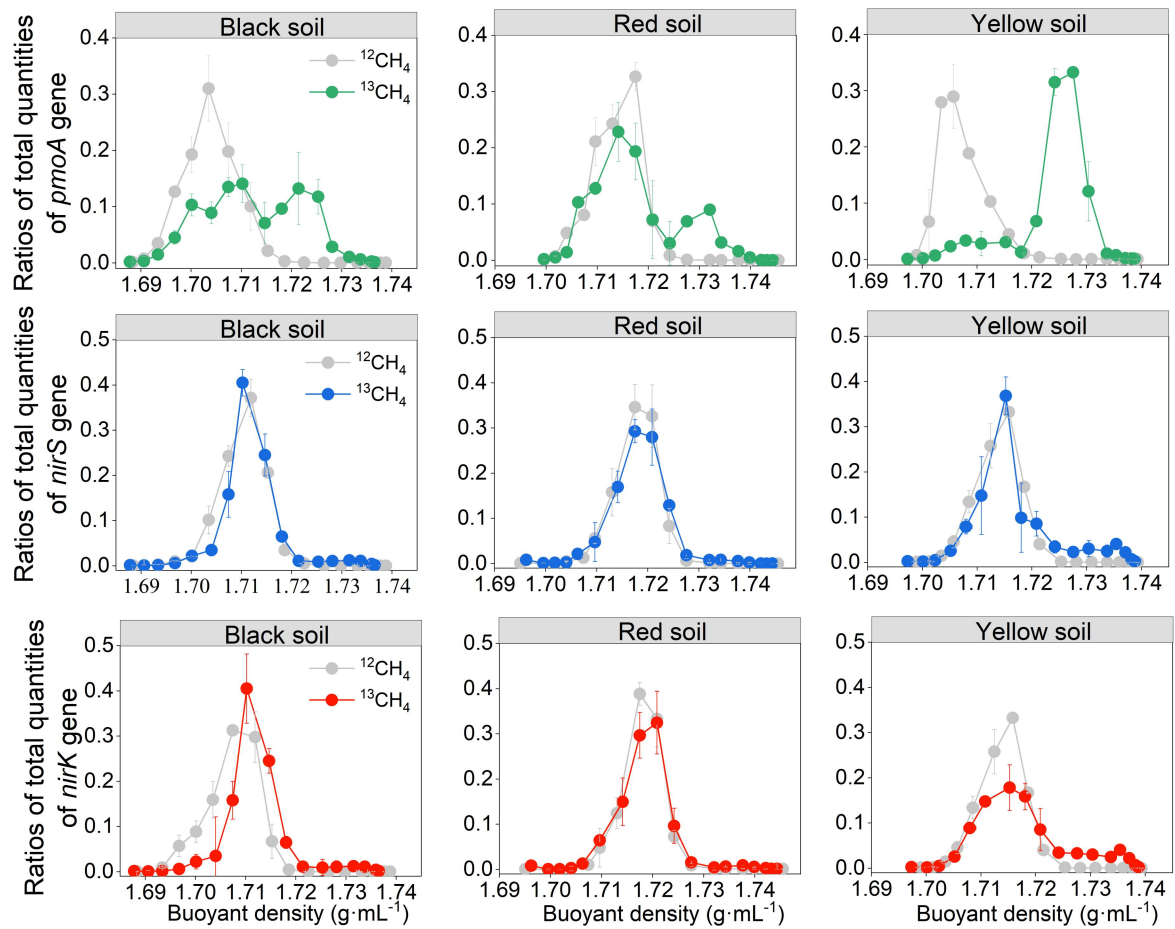
Supplementary Fig. 8 Effects of methane (CH₄) addition on microbial CH₄ oxidation of three typical paddy soils. A-B Variations in CH₄ and carbon dioxide (CO₂) emissions; **C-D** Changes in *pmoA* gene transcription and key genera of methanotrophs using RNA reverse transcription in CH₄ addition experiments. The error bar in (A-C) represents the standard error of triplicate samples, and data are presented as mean values ± standard error. Different lowercase letters in (C) indicate significant differences between the soils with different CH₄ concentrations ($p < 0.05$; $n = 3$; one-way ANOVA followed by two-sided Tukey post hoc test). * indicates statistically significant levels of $p < 0.05$ and ** indicates $p < 0.01$ based on two-sided T-Test in (D). Exact p -values and Source data are provided as a Source Data file.



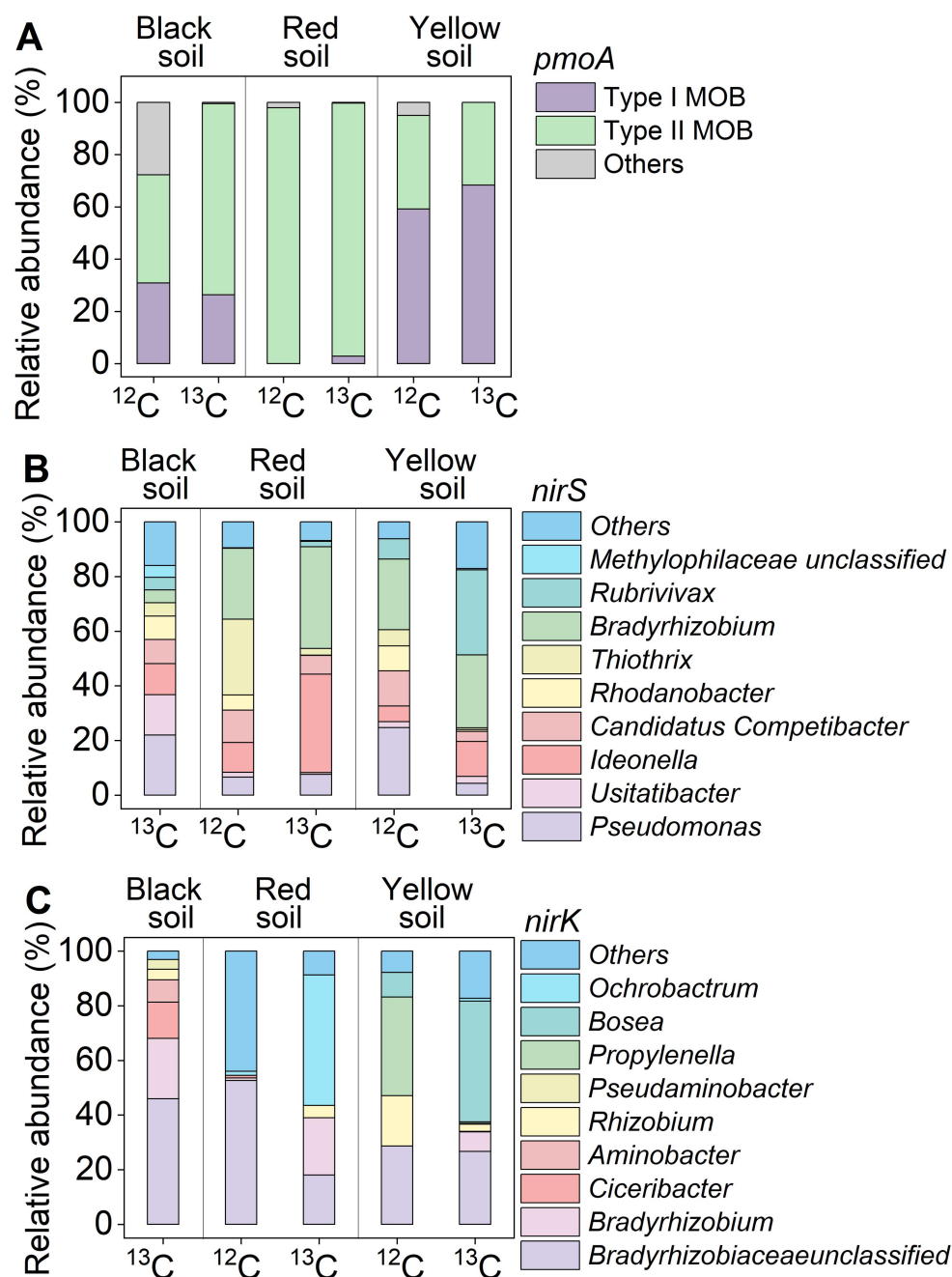
Supplementary Fig. 9 Effects of addition of methane (CH₄) and methanotrophs on the composition of functional genes involved in CH₄ oxidation and denitrification. A The effects of CH₄ addition on bacterial beta-diversity in CH₄ oxidizing gene (*pmoA*); **B** Methanotrophs addition affects bacterial beta-diversity of denitrification genes (*nirS* and *nirK*) by permutational multivariate analysis of variance (PERMANOVA). Source data are provided as a Source Data file.



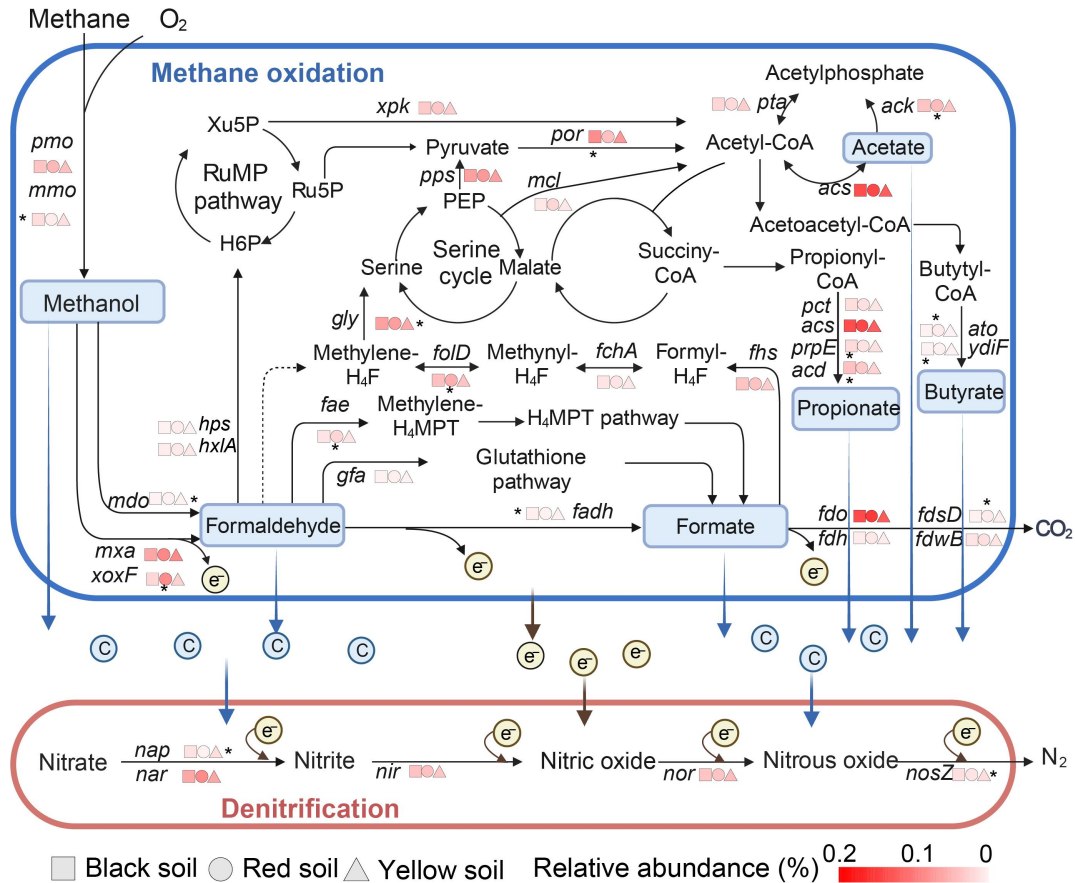
Supplementary Fig. 10 Effects of addition of aerobic methanotrophs on soil denitrification. **A** Variations in consumption of nitrate (NO₃⁻-N); **B** Changes in the transcript of denitrification genes (*narG*, *norB*, *nosZI* and *nosZII*) based on RNA reverse transcription. The error bar represents the standard error of triplicate samples, and data are presented as mean values ± standard error. Different lowercase letters indicate significant differences between soils with different amounts of aerobic methanotrophs added ($p < 0.05$; $n = 3$; one-way ANOVA followed by two-sided Tukey post hoc test). Exact p -values and Source data are provided as a Source Data file.



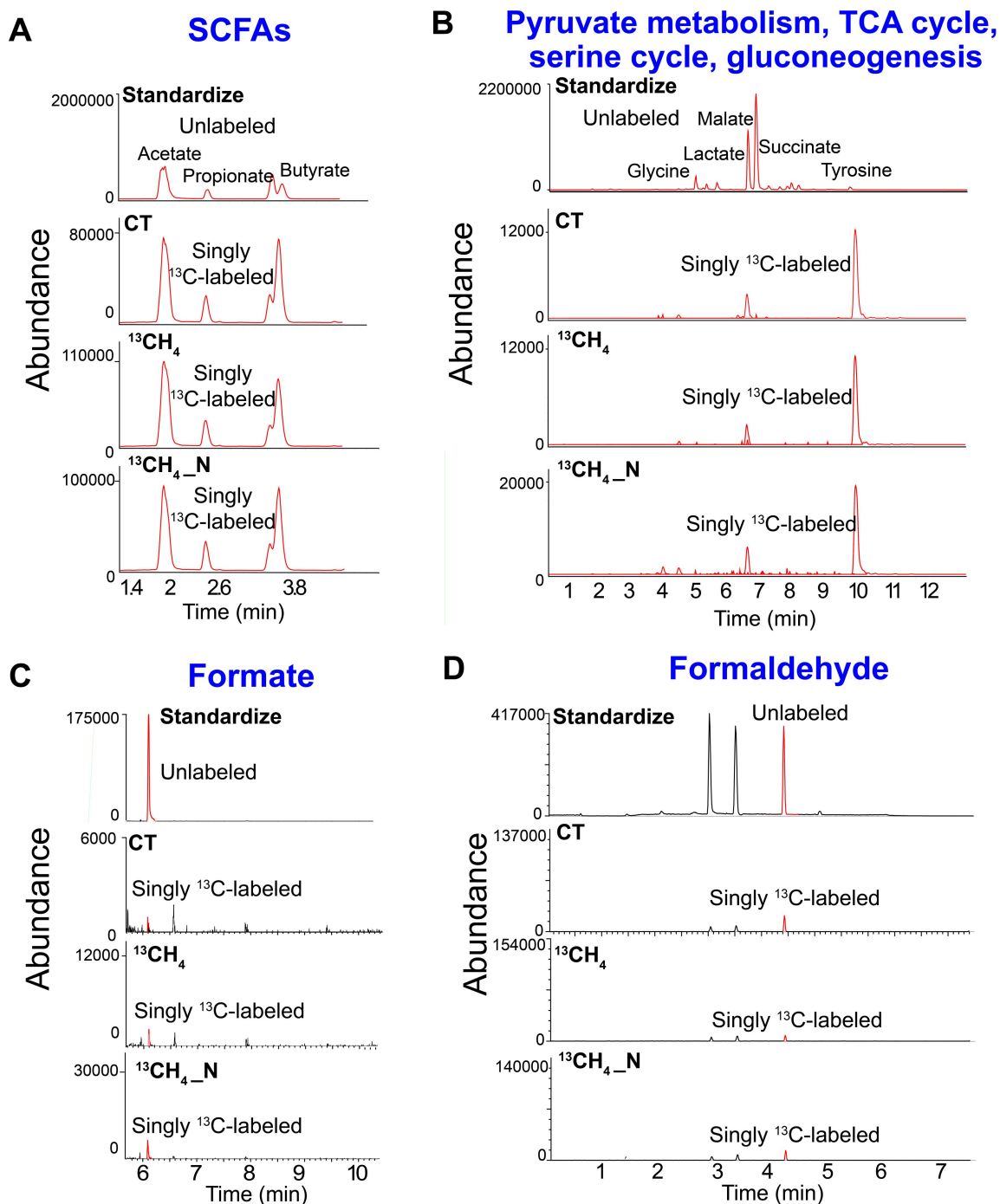
Supplementary Fig. 11 Quantitative distribution of genes involved in methane (CH₄) oxidation (*pmoA*) and denitrification (*nirS* and *nirK*) in eighteen fractions from each CsCl gradient in the three soils. Quantitative distribution of the *pmoA*, *nirS* and *nirK* genes in eighteen fractions from each CsCl gradient, covering a density range from 1.69 to 1.75 g mL⁻¹ of the DNA fractions from three typical soils incubated with ¹²CH₄ and ¹³CH₄ for 15 days. The error bar represents the standard error of triplicate samples, and data are presented as mean values ± standard error. Source data are provided as a Source Data file.



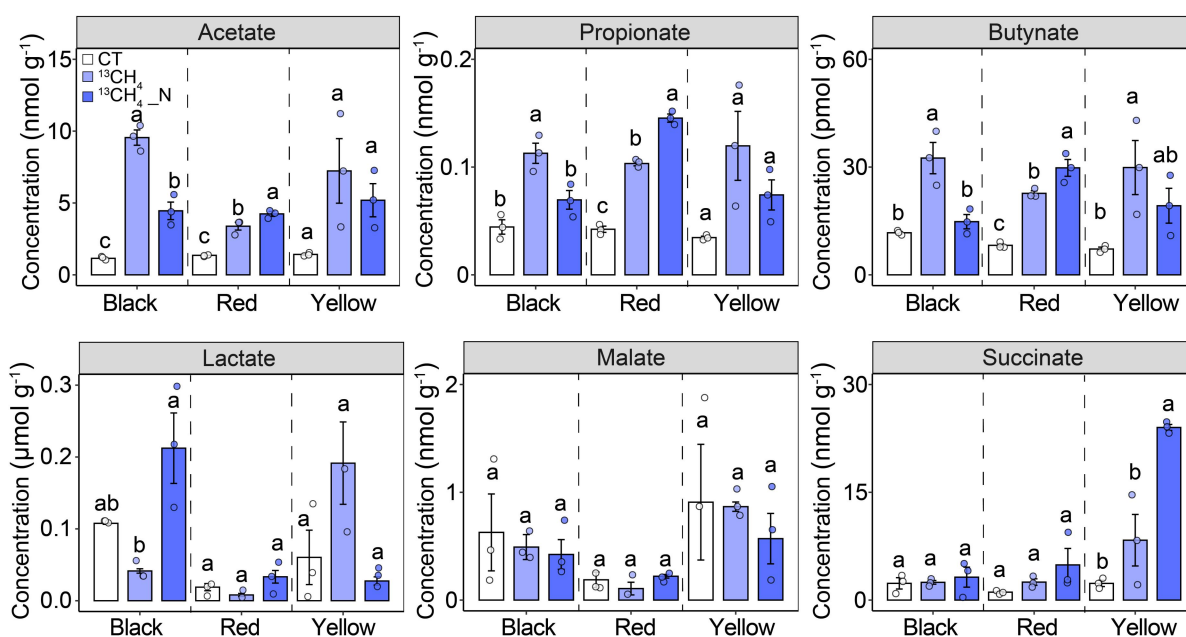
99 **Supplementary Fig. 12 Community composition of methanotrophs and denitrifiers in**
 100 **the ^{12}C -methane (CH_4) and ^{13}C - CH_4 treatments in three typical soils. A** Dominant types
 101 **of methanotrophs in the ^{12}C - CH_4 and ^{13}C - CH_4 treatments in three typical soils; B-C** Relative
 102 **abundance of the top ten genera of *nirS*-denitrifiers and *nirK*-denitrifiers in the ^{12}C - CH_4 and**
 103 **^{13}C - CH_4 treatments. Source data are provided as a Source Data file.**



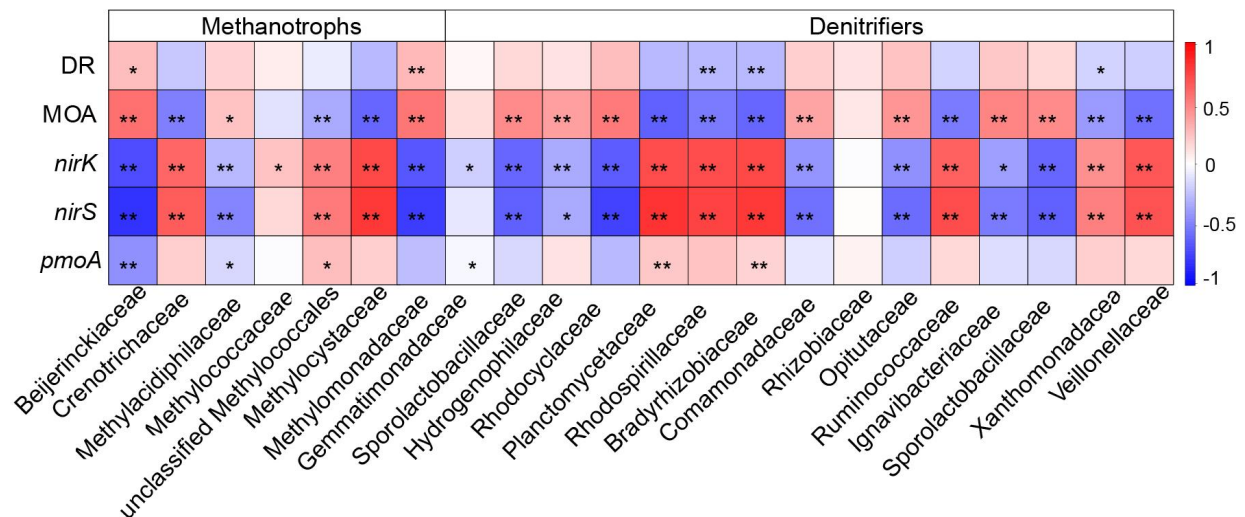
Supplementary Fig. 13 The proposed metagenomic pathways of the coupling between aerobic methane (CH_4) oxidation and denitrification at the community level in the paddy soils. The color gradient represents the relative abundances of major genes of metagenomic analysis in the heavy DNA from $^{13}CH_4$ incubation. * indicates statistically significant differences in the three selected soils at levels of $p < 0.05$ based on one-way ANOVA followed by two-sided Tukey post hoc test. The square, circle, and triangle represent genes in black soil, red soil, and yellow soil, respectively. The definitions of abbreviations are listed in Supplementary Table 6. Source data are provided as a Source Data file.



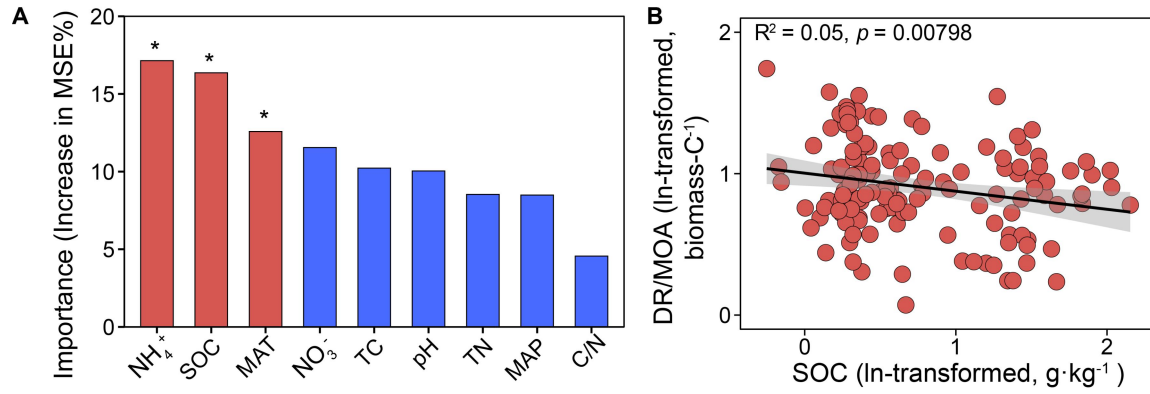
Supplementary Fig. 15 Typical LC-MS/MS and GC-MS chromatograms of targeted metabolites originating from methane (CH_4) oxidation in three typical soils. A Short chain fatty acids (SCFAs); **B** The intermediates involved in pyruvate metabolism, TCA cycle, serine cycle and gluconeogenesis; **C** Formate; **D** Formaldehyde.



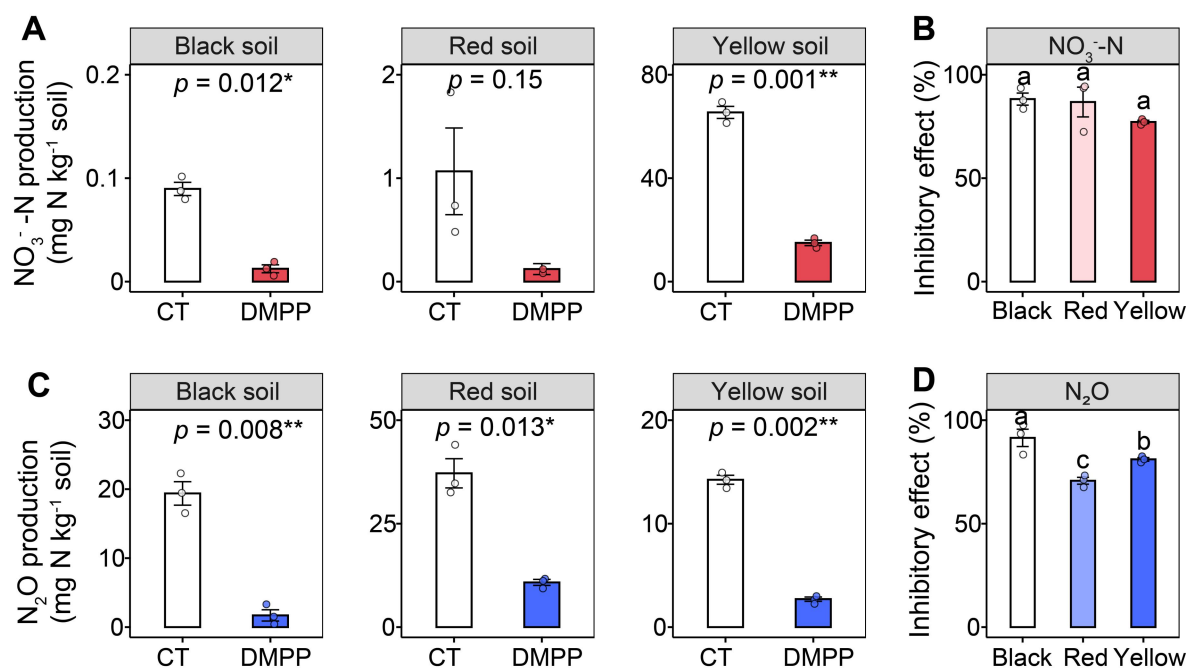
Supplementary Fig. 16 The concentration of metabolites originating from methane (CH_4) oxidation in three typical soils. The error bar in (B) represents the standard error of triplicate samples, and data are presented as mean values $\pm$ standard error ($n = 3$; one-way ANOVA followed by two-sided Tukey post hoc test). Different lowercase letters in (B) indicate significant differences between treatments ($p < 0.05$). Exact p -values and Source data are provided as a Source Data file.



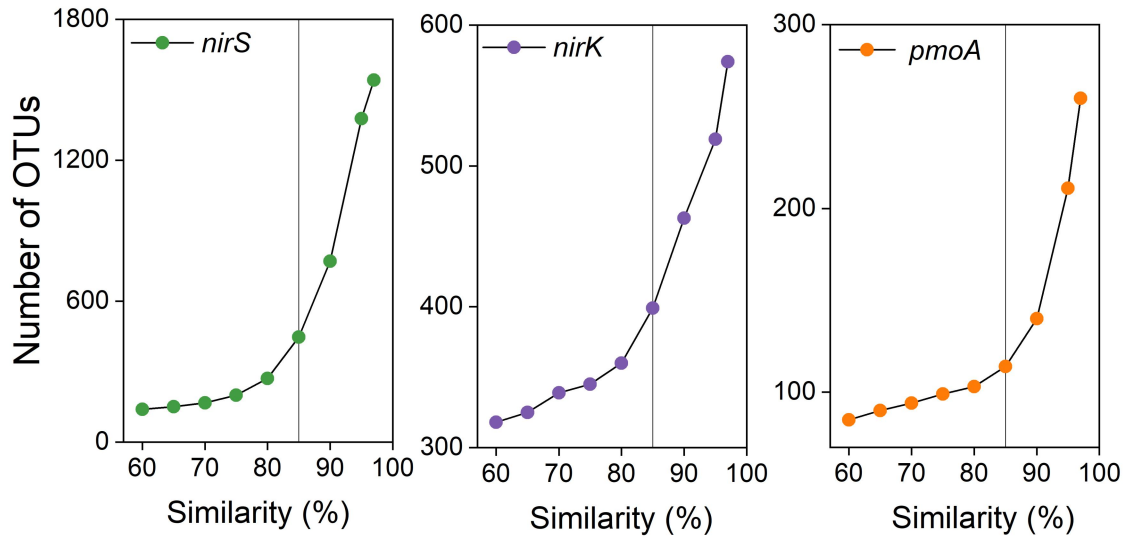
Supplementary Fig. 17 Pearson correlations of activities and genes with key taxa related to methane (CH₄) oxidation and denitrification. DR, denitrification rate; MOA, CH₄ oxidizing activity. Significant correlations are indicated by *. * indicated $p < 0.05$ and ** indicated $p < 0.01$. Source data are provided as a Source Data file.



Supplementary Fig. 18 Factors predicting the relationships between methane oxidation and denitrification. **A** Random forest analysis identifying significant predictors on the relative denitrification to methane oxidation activities (DR/MBC)/ (MOA/MBC) ($p < 0.05$); **B** The relationships between (DR/MBC)/ (MOA/MBC) and SOC. SOC, soil organic carbon; MAT, mean annual temperature; MAP, mean annual precipitation. Exact p -values in (A) and Source data are provided as a Source Data file.



Supplementary Fig. 19 The inhibition effects of 3,4-dimethylpyrazole phosphate (DMPP) on nitrification in the three soils. **A** Variations in production of nitrate (NO_3^-); **B** The inhibitory effects of DMPP on NO_3^- production; **C** Changes in nitrous oxide (N_2O) emissions; **D** The inhibitory effects of DMPP on N_2O emissions. The error bar represents the standard error of triplicate samples, and data are presented as mean values $\pm$ standard error. * indicates statistically significant levels of $p < 0.05$ and ** indicates $p < 0.01$ based on two-sided T-Test in (A) and (C). Different lowercase letters in (B) and (D) indicate significant differences between treatments ($p < 0.05$; one-way ANOVA followed by two-sided Tukey post hoc test). Exact p -values and Source data are provided as a Source Data file.



153 **Supplementary Fig. 20 The counts of operational taxonomic units (OTUs) clustering at**
 154 **various thresholds. Source data are provided as a Source Data file.**

Supplementary Table 1. Geographical information and physicochemical characteristics of collected natural soil samples.

Sample	Longitude (°)	Latitude (°)	TC (%)	TN (%)	pH	Clay (g kg ⁻¹)	SOC (g kg ⁻¹)
Black soil	126.63	47.43	3.76	0.28	6.43	334	37.53
Red soil	112.83	28.56	2.26	0.21	4.69	227	23.08
Yellow soil	113.36	23.16	1.31	0.12	5.68	192	14.10

Supplementary Table 2. Properties of recovered metagenome assembled genomes (MAGs) affiliated with methanotrophs and denitrifiers with completeness >50% and contamination <10%.

Bin Id	Contigs	Genome size (bp)	N50	Completeness (%)	Contamination (%)	16S rRNA	23S rRNA	5S rRNA	tRNA	Bin quality	Abundance (%)
B_Bin42	298	5242199	27616	98.94	1.63	0	0	1	40	MQ	1.36
B_Bin35	145	4715769	73841	98.21	2.13	0	0	0	47	MQ	2.25
B_Bin44	708	5957189	12457	95.36	7.94	0	0	0	42	MQ	0.99
B_Bin46	200	5283043	60611	94.70	4.87	0	0	0	46	MQ	0.57
B_Bin51	257	4855676	31935	97.30	5.07	0	0	1	48	MQ	0.89
B_Bin11	1087	6039704	7597	91.16	1.02	0	0	0	40	MQ	0.43
B_Bin75	835	5249229	8093	90.31	3.89	0	1	1	45	MQ	0.30
B_Bin70	189	2931300	28219	83.10	0.86	0	0	0	41	MQ	0.64
B_Bin82	751	3024827	4681	82.91	1.42	0	0	0	30	MQ	0.28
R_Bin35	160	3431118	120546	97.44	2.27	0	0	0	45	MQ	0.69
R_Bin56	967	7028512	144445	94.68	8.73	0	0	0	63	MQ	0.54
R_Bin14	504	3899330	52019	84.69	2.48	0	0	0	32	MQ	5.15
R_Bin29	781	3403938	37244	83.75	3.55	0	0	0	35	MQ	1.29
R_Bin61	186	3452242	114161	71.23	3.51	0	0	0	26	MQ	5.31
R_Bin37	82	2444772	183383	68.94	2.61	0	0	1	33	MQ	2.03
R_Bin52	48	3116807	262841	98.72	0.11	0	0	0	40	MQ	4.71
R_Bin10	27	2316567	396376	98.29	0.47	0	0	1	41	MQ	0.99
R_Bin2	390	3872137	58474	95.48	0.89	0	0	0	32	MQ	0.64
R_Bin38	974	2087582	6531	52.15	6.29	0	0	0	5	MQ	0.91
Y_Bin93	304	4721736	29398	87.83	7.29	0	0	0	35	MQ	9.07
Y_Bin80	127	3945945	50637	75.31	1.08	0	0	0	38	MQ	3.37
Y_Bin97	1031	3227355	3397	69.66	3.11	0	0	0	35	MQ	1.56
Y_Bin89	44	5674177	202138	99.32	3.42	0	0	1	44	MQ	1.89
Y_Bin87	103	3485676	53718	99.18	1.48	0	0	1	42	MQ	1.26
Y_Bin47	49	2998556	152796	98.41	2.30	0	0	0	45	MQ	0.92
Y_Bin19	336	4254566	23528	96.40	1.55	0	0	0	52	MQ	1.21
Y_Bin41	425	4084344	16372	95.73	4.50	0	0	0	43	MQ	0.45
Y_Bin63	117	4337290	114236	94.54	5.29	0	0	0	45	MQ	1.08
Y_Bin11	734	2187264	3226	98.81	5.02	0	0	1	35	MQ	0.49
Y_Bin1	304	4721736	29398	62.99	4.98	0	0	0	17	MQ	0.53

162 **Supplementary Table 3.** The determined various intermediates involved in CH₄ oxidation.

Pathway	Metabolite	Formula	Exact mass
C1	Formaldehyde	CH ₂ O	30.0106
	Formate	CH ₂ O ₂	46.0055
Short-chain fatty acids	Acetate	C ₂ H ₄ O ₂	60.0211
	Propionate	C ₃ H ₆ O ₂	74.0368
	Butyrate	C ₄ H ₈ O ₂	88.0524
TCA cycle and pyruvate metabolism	Citrate	C ₆ H ₈ O ₇	192.0270
	Succinate	C ₄ H ₆ O ₄	118.0266
	Malate	C ₄ H ₆ O ₅	134.0215
	Oxalate	C ₂ H ₂ O ₄	89.9953
	Lactate	C ₃ H ₆ O ₃	90.0317
	Oxaloacetate	C ₄ H ₄ O ₅	132.0059
	Fumarate	C ₄ H ₄ O ₄	116.0110
Serine cycle	Pyruvate	C ₃ H ₄ O ₃	88.0160
	Glycine	C ₂ H ₅ NO ₂	75.0320
	Serine	C ₃ H ₇ NO ₃	105.0426
	Phosphoenolpyruvate	C ₃ H ₅ O ₆ P	167.9824
	Tyrosine	C ₉ H ₁₁ NO ₃	181.0739
	Hydroxypyruvate	C ₃ H ₄ O ₄	104.0110
	D-Glycerate	C ₃ H ₆ O ₄	106.0266
	Glutamine	C ₅ H ₁₀ N ₂ O ₃	146.0691
	Glutamate	C ₅ H ₉ NO ₄	147.0532
	Alanine	C ₃ H ₇ NO ₂	89.0477
Gluconeogenesis	Fructose 6-phosphate	C ₆ H ₁₃ O ₉ P	260.0297
	Glyceraldehyde 3-phosphate	C ₃ H ₇ O ₆ P	169.9980
	Ribulose 5-phosphate	C ₅ H ₁₁ O ₈ P	230.0192
	Acetyl phosphate	C ₂ H ₅ O ₅ P	139.9875
	Xylulose 5-phosphate	C ₅ H ₁₁ O ₈ P	230.0192
	Fructose 1,6-bisphosphate	C ₆ H ₁₄ O ₁₂ P ₂	339.9960
	2-Phospho-D-glycerate	C ₃ H ₇ O ₇ P	185.9929
	Glucose	C ₆ H ₁₂ O ₆	180.0634

163

Supplementary Table 4. Primer sets and amplification programs used in the quantitative PCR.

Gene	Primer	Sequence (5'-3')	Protocol	Ref
<i>pmoA</i>	A189f	GGNGACTGGGACTTCTGG	95°C/3min, 35 cycles	1, 2
	mb661r	CCGGMGCAACGTCYTTACC	(94°C/45s, 53°C/60s, 72°C/2min), 72°C/4min	
<i>nirK</i>	1F	GGMATGGTKCCSTGGCA	95°C/3min, 40 cycles	3
	5R	GCCTCGATCAGRTRTGG	(95°C/30s, 59°C/40s, 72°C/30s), 72°C/10min	
<i>nirS</i>	cd3aF	G TSAACG TSAAGGARACSGG	95°C/3min, 30 cycles	4
	R3cd	GASTTCGGRTGSGTCTTGA	(95°C/45s, 57°C/45s, 72°C/45s), 72°C/5min	
<i>narG</i>	1960m2F	TAYGTSGGGCAGGARAACTG	95°C/3 min, 35 cycles	5
	2050m2R	CGTAGAAGAAGCTGGTGCTGTT	(95°C/10s, 56°C/30s, 72°C/20s), 72°C/10min	
<i>norB</i>	qnorB2F	GGNCAYCARGGNTAYGA	95°C/3min, 40 cycles	6
	qnorB5R	ACCCANAGRTGNACNACCCACCA	(94°C/15s, 54°C/30s, 72°C/45s), 72°C/10min	
<i>nosZI</i>	2F	CGCRACGGCAASAAGGTSMSSGT	95°C/3min, 35 cycles	7
	2R	CAKRTGCAKSGCRTGGCAGAA	(95°C/45s, 57°C/45s, 72°C/45s), 72°C/5min	
<i>nosZII</i>	1F	CTIGGICCIYTKCAYAC	95°C/3min, 40 cycles	8
	2R	GCIGARCARAAITCBGTRC	(95°C/15s, 54°C/30s, 72°C/30s), 72°C/30s	

167 **Supplementary Table 5.** Denitrifiers identified across paddy fields of China.

Denitrifiers	Survival condition	Ref	Denitrifiers	Survival condition	Ref
<i>Caldilineaceae</i>	anaerobic	9	<i>Phycisphaeraceae</i>	anaerobic	10
<i>Cellulomonadaceae</i>	facultatively anaerobic	11	<i>Ignavibacteriaceae</i>	anaerobic	12
<i>Helicobacteraceae</i>	microaerobic	13	<i>Gracilibacteria_fa</i>	anaerobic	14
<i>Polyangiaceae</i>	aerobic	15	<i>Veillonellaceae</i>	anaerobic	16
<i>Bdellovibrionaceae</i>	aerobic	17	<i>Ruminococcaceae</i>	anaerobic	18
<i>Methylophilaceae</i>	aerobic	19	<i>Peptococcaceae</i>	anaerobic	20
<i>Alcaligenaceae</i>	aerobic	21	<i>Lachnospiraceae</i>	anaerobic	22
<i>Rhodobacteraceae</i>	aerobic	23	<i>Clostridiaceae_1</i>	anaerobic	24
<i>Xanthobacteraceae</i>	aerobic	25	<i>JG30-KF-CM45_fa</i>	anaerobic	26
<i>Rhizobiaceae</i>	aerobic	25	<i>HSB_OF53-F07</i>	anaerobic	27
<i>Cyclobacteriaceae</i>	aerobic	28	<i>Chloroflexi_fa</i>	anaerobic	26
<i>Gaiellaceae</i>	aerobic	29	<i>Frankiaceae</i>	anaerobic	30
<i>Euzebyaceae</i>	aerobic	31	<i>Rhodospirillales_unclassified</i>	anaerobic	32
<i>Mycobacteriaceae</i>	aerobic	33	<i>Hyphomicrobiaceae</i>	anaerobic	34
<i>Hyphomonadaceae</i>	aerobic	35	<i>Gemmatimonadaceae</i>	facultatively aerobic	36
<i>Sandaracinaceae</i>	anaerobic	37	<i>Chloroflexaceae</i>	facultatively aerobic	38
<i>Gallionellaceae</i>	anaerobic	39	<i>Rhodocyclaceae</i>	facultatively anaerobic	40
<i>Oxalobacteraceae</i>	anaerobic	41	<i>Hydrogenophilaceae</i>	facultatively anaerobic	42
<i>Comamonadaceae</i>	anaerobic	43	<i>Caulobacteraceae</i>	facultatively anaerobic	11
<i>Phyllobacteriaceae</i>	anaerobic	44	<i>Bacillaceae</i>	facultatively anaerobic	45
<i>Bradyrhizobiaceae</i>	anaerobic	46	<i>Flavobacteriaceae</i>	facultatively anaerobic	47
<i>Planctomycetaceae</i>	anaerobic	48	<i>Cytophagaceae</i>	facultatively anaerobic	49

Supplementary Table 6. The associations of climate and soil with microbial attributes based on the structural equation modeling.

Parameters			Standardized regression weights	Regression weights	<i>P</i>
MAT	→	<i>nirK</i>	0.229	0.092	0.037
MAT	→	<i>nirS</i>	0.264	0.207	0.004
MAT	→	<i>pmoA</i>	0.253	0.079	0.048
MAP	→	<i>nirK</i>	-0.371	-1.879	< 0.001
MAP	→	<i>nirS</i>	-0.203	-1.998	0.031
MAP	→	<i>pmoA</i>	-0.322	-1.253	0.014
SOC	→	<i>nirK</i>	0.457	1.007	< 0.001
SOC	→	<i>nirS</i>	0.527	2.258	< 0.001
SOC	→	<i>pmoA</i>	0.343	0.581	< 0.001
pH	→	<i>pmoA</i>	0.227	1.637	0.023
NH ₄ ⁺	→	<i>nirS</i>	-0.281	-0.770	< 0.001
NH ₄ ⁺	→	NMDS1	0.235	0.193	0.028
pH	→	NMDS1	0.398	2.180	< 0.001

Supplementary Table 7. The description of genes involved in methane oxidation and denitrification.

Genes	Description	Genes	Description
<i>pmo</i>	particulate methane monooxygenase	<i>mmo</i>	soluble methane monooxygenase
<i>edd</i>	phosphogluconate dehydratase	<i>xpk</i>	xylulose-5-phosphate phosphoketolase
<i>mxs</i>	calcium-dependent methanol dehydrogenase	<i>xoxF</i>	lanthanide-dependent methanol dehydrogenase
<i>fae</i>	formaldehyde activating enzyme	<i>mtd</i>	methylene tetrahydromethanopterin
<i>mch</i>	methenyl tetrahydromethanopterin cyclohydrolase	<i>hxlA</i>	3-hexulose-6-phosphate synthase
<i>hps</i>	3-hexulose-6-phosphate synthase	<i>hxlB</i>	6-phospho-3-hexuloisomerase
<i>pfk</i>	6-phosphofructokinase	<i>aldo</i>	fructose-bisphosphate aldolase
<i>ftr</i>	formylmethanofuran-tetrahydromethanopterin	<i>mdo</i>	formaldehyde dismutase / methanol dehydrogenase
<i>fdh</i>	NAD-dependent formate dehydrogenase	<i>fdo, fdsD, fdw</i>	formate dehydrogenase major subunit
<i>gly</i>	serine hydroxymethyltransferase	<i>agxt</i>	serine-glyoxylate aminotransferase
<i>hpr</i>	hydroxypyruvate reductase	<i>gck</i>	glycerate kinase
<i>eno</i>	enolase	<i>ppc</i>	phosphoenolpyruvate carboxylase
<i>madh</i>	malate dehydrogenase	<i>mtk</i>	malate-CoA ligase
<i>mcl</i>	malyl-CoA lyase	<i>fum</i>	fumarate hydratase
<i>sdh</i>	succinate dehydrogenase	<i>suc</i>	succinyl-CoA synthetase
<i>acnA</i>	aconitate hydratase	<i>cs</i>	citrate synthase
<i>pps</i>	pyruvate, water dikinase	<i>por</i>	pyruvate ferredoxin oxidoreductase
<i>pta</i>	phosphate acetyltransferase	<i>ack</i>	acetate kinase
<i>acs</i>	acetyl-CoA synthetase	<i>gfa</i>	S-(hydroxymethyl)glutathione synthase
<i>fadh</i>	glutathione-independent formaldehyde dehydrogenase	<i>folD</i>	methylenetetrahydrofolate dehydrogenase
<i>fchA</i>	methenyltetrahydrofolate cyclohydrolase	<i>fhs</i>	formate--tetrahydrofolate ligase
<i>tkt</i>	transketolase	<i>tal</i>	transaldolase
<i>rpe</i>	ribulose-phosphate 3-epimerase	<i>gnt</i>	6-phosphogluconate dehydrogenase
<i>ato</i>	acetate CoA -transferase	<i>pct</i>	propionate CoA-transferase
<i>nar/nap</i>	periplasmic nitrate reductase	<i>nir</i>	copper-containing nitrite reductase
<i>nor</i>	nitric oxide reductase	<i>nosZ</i>	nitrous-oxide reductase

Supplementary References

1. Knief, C., Lipski, A. & Dunfield, P. F., Diversity and activity of methanotrophic bacteria in different upland soils. *Appl. Environ. Microbiol.* **69**, 6703-6714 (2003).
2. Costello, A. M. & Lidstrom, M. E., Molecular characterization of functional and phylogenetic genes from natural populations of methanotrophs in lake sediments. *Appl. Environ. Microbiol.* **65**, 5066-5074 (1999).
3. Braker, G., Fesefeldt, A. & Witzel, K.-P., Development of PCR primer systems for amplification of nitrite reductase genes (nirK and nirS) to detect denitrifying bacteria in environmental samples. *Appl. Environ. Microbiol.* **64**, 3769-3775 (1998).
4. Throbäck, I. N., Enwall, K., Jarvis, Å. & Hallin, S., Reassessing pcr primers targeting nirS, nirK and nosZ genes for community surveys of denitrifying bacteria with DGGE. *FEMS. Microbiol. Ecol.* **49**, 401-417 (2004).
5. Kandeler, E., Deiglmayr, K., Tschierko, D., Bru, D. & Philippot, L., Abundance of narG, nirS, nirK, and nosZ genes of denitrifying bacteria during primary successions of a glacier foreland. *Appl. Environ. Microbiol.* **72**, 5957-5962 (2006).
6. Braker, G. & Tiedje, J. M., Nitric oxide reductase (norB) genes from pure cultures and environmental samples. *Appl. Environ. Microbiol.* **69**, 3476-3483 (2003).
7. Henry, S., Bru, D., Stres, B., Hallet, S. & Philippot, L., Quantitative detection of the nosZ gene, encoding nitrous oxide reductase, and comparison of the abundances of 16S rRNA, narG, nirK, and nosZ genes in soils. *Appl. Environ. Microbiol.* **72**, 5181-5189 (2006).
8. Jones, C. M., Graf, D. R., Bru, D., Philippot, L. & Hallin, S., The unaccounted yet abundant nitrous oxide-reducing microbial community: A potential nitrous oxide sink. *ISME. J.* **7**, 417-426 (2013).
9. Wu, L., Li, Z., Zhao, C., Liang, D. & Peng, Y. A novel partial-denitrification strategy for post-anammox to effectively remove nitrogen from landfill leachate. *Sci. Total. Environ.* **633**, 745-751 (2018).
10. Jasmin, C. et al. Diversity of sediment-associated planctomycetes in the arabian sea oxygen minimum zone. *J. Basic. Microbiol.* **57**, 1010-1017 (2017).
11. Garrity, G. M., Bell, J. A. & Lilburn, T. Caulobacteraceae. *Bergey's Manual of Systematics of Archaea and Bacteria* (Springer: Berlin, Heidelberg, 2015), pp. 1-25.
12. Zheng, Y. et al. Identifying microbial community evolution in membrane bioreactors coupled with anaerobic side-stream reactor, packing carriers and ultrasonication for sludge reduction by linear discriminant analysis. *Bioresource. Technol.* **291**, 121920 (2019).

- 208 13. Garrity, G. M., Bell, J. A. & Lilburn, T. Helicobacteraceae fam. Nov. *Bergey's Manual*
209 *of Systematics of Archaea and Bacteria* (Springer: Berlin, Heidelberg, 2014), pp. 1-1
210 (2015).
- 211 14. Ji, J. et al. Effects of salinity build-up on the performance and microbial community
212 of partial-denitrification granular sludge with high nitrite accumulation. *Chemosphere*
213 **209**, 53-60 (2018).
- 214 15. Garcia, R. & Müller, R. The family polyangiaceae. *Bergey's Manual of Systematics of*
215 *Archaea and Bacteria* (Springer: Berlin, Heidelberg, 2014), pp. 247-279.
- 216 16. Carlier, J.-P. et al. Anaeroglobus geminatus gen. Nov., sp. Nov., a novel member of
217 the family veillonellaceae. *Int. J. Syst. Evol. Microbiol.* **52**, 983-986 (2002).
- 218 17. Sockett, R. E. & Lambert, C. Bdellovibrio as therapeutic agents: A predatory
219 renaissance? *Nat. Rev. Microbiol.* **2**, 669-675 (2004).
- 220 18. Rainey, F. A. Family VIII. Ruminococcaceae fam. Nov. *Bergey's manual of systematic*
221 *bacteriology* (Springer: Berlin, Heidelberg, 2009), pp.1016-1043.
- 222 19. Osaka, T. et al. Identification of acetate-or methanol-assimilating bacteria under
223 nitrate-reducing conditions by stable-isotope probing. *Microb. Ecol.* **52**, 253-266
224 (2006).
- 225 20. Atashgahi, S. et al. A benzene-degrading nitrate-reducing microbial consortium
226 displays aerobic and anaerobic benzene degradation pathways. *Sci. Rep.* **8**, 1-12
227 (2018).
- 228 21. Wang, H. et al. Breast tissue, oral and urinary microbiomes in breast cancer.
229 *Oncotarget* **8**, 88122 (2017).
- 230 22. Rainey, F. A. Family V. Lachnospiraceae fam. Nov. *Bergey's manual of systematic*
231 *bacteriology* (Springer: Berlin, Heidelberg, 2009), pp. 921-968 (2009).
- 232 23. Pujalte, M. J., Lucena, T., Ruvira, M. A., Arahal, D. R. & Macián, M. C. The family
233 rhodobacteraceae. (Springer, 2014), pp. 439-512. (2014).
- 234 24. Heller, H. H. Certain genera of the clostridiaceae: Studies in pathogenic anaerobes. V.
235 *J. Bacteriol.* **7**, 1-36 (1922).
- 236 25. Alves, L. M. C., De Souza, J. A. M., de Mello Varani, A. & de Macedo Lemos, E. The
237 family rhizobiaceae. *The Prokaryotes*, 419-437 (Springer: Berlin, Heidelberg, 2014),
238 pp. 419-437. (2014).
- 239 26. Nunoura, T. et al. Isolation and characterization of a thermophilic, obligately
240 anaerobic and heterotrophic marine chloroflexi bacterium from a
241 chloroflexi-dominated microbial community associated with a japanese shallow
242 hydrothermal system, and proposal for thermomarinilinea lacunofontalis gen. Nov., sp.

- 243 Nov. *Microbes. Environ.* **28**, 228-235 (2013).
- 244 27. Lezcano, M. Á. et al. Biomarker profiling of microbial mats in the geothermal band of
245 cerro caliente, deception island (antarctica): Life at the edge of heat and cold.
246 *Astrobiology* **19**, 1490-1504 (2019).
- 247 28. Kumar, A. et al. Shivajiella indica gen. Nov., sp. Nov., a marine bacterium of the
248 family “cyclobacteriaceae” with nitrate reducing activity. *Syst. Appl. Microbiol.* **35**,
249 320-325 (2012).
- 250 29. da Costa, L. A. M. S. 19 The family Gaiellaceae. *The. Prokaryotes-Actinobacteria*
251 (Springer: Berlin, Heidelberg, 2014).
- 252 30. Woods, D. R. & Reid, S. J. Recent developments on the regulation and structure of
253 glutamine synthetase enzymes from selected bacterial groups. *FEMS. Microbiol. Rev.*
254 **11**, 273-283 (1993).
- 255 31. Xu, L. et al. Complete genome sequence of euzebya sp. Dy32-46, a marine
256 actinobacteria isolated from the pacific ocean. *Mar. Genom.* **44**, 65-69 (2019).
- 257 32. López-López, A. et al. Extremely halophilic microbial communities in anaerobic
258 sediments from a solar saltern. *Environ. Microbiol. Rep.* **2**, 258-271 (2010).
- 259 33. Magee, J. G. & Ward A. C. Family III. Mycobacteriaceae chester 1897, 63al. *Bergey's*
260 *Manual of Systematic Bacteriology: Volume 5: The Actinobacteria* (Springer: Berlin,
261 Heidelberg, 2012), pp. 312. (2012).
- 262 34. Oren, A. & Xu, X.-W. The family Hyphomicrobiaceae. *The Prokaryotes:*
263 *Alphaproteobacteria and Betaproteobacteria* (Springer Berlin Heidelberg, Berlin,
264 Heidelberg, 2014), pp. 247-281. (2014).
- 265 35. Lee, K., Lee, H. K., Choi, T.-H. & Cho, J.-C. Robiginitomaculum antarcticum gen.
266 Nov., sp. Nov., a member of the family hyphomonadaceae, from antarctic seawater.
267 *Int. J. Syst. Evol. Microbiol.* **57**, 2595-2599 (2007).
- 268 36. Hanada, S. & Sekiguchi, Y. in *The prokaryotes: Other major lineages of bacteria and*
269 *the archaea*, E. Rosenberg, E. F. DeLong, S. Lory, E. Stackebrandt, F. Thompson, Eds.
270 (Springer Berlin Heidelberg, Berlin, Heidelberg, 2014), pp. 677-681.
- 271 37. Mohr, K. I., Garcia, R. O., Gerth, K., Irschik, H. & Müller, R. Sandaracinus
272 amylolyticus gen. Nov., sp. Nov., a starch-degrading soil myxobacterium, and
273 description of sandaracinaceae fam. Nov. *Int. J. Syst. Evol. Microbiol.* **62**, 1191-1198
274 (2012).
- 275 38. Hanada, S. & Pierson, B. K. The family chloroflexaceae. *The prokaryotes* (Springer:
276 Berlin, Heidelberg, 2006), pp. 815-842. (2006).
- 277 39. Bryce, C. et al. Microbial anaerobic Fe (II) oxidation—ecology, mechanisms and

environmental implications. *Environ. Microbiol.* **20**, 3462-3483 (2018).

40. Oren, A. in *The prokaryotes: Alphaproteobacteria and betaproteobacteria*, E. Rosenberg, E. F. DeLong, S. Lory, E. Stackebrandt, F. Thompson, Eds. (Springer Berlin Heidelberg, Berlin, Heidelberg, 2014), pp. 975-998.
41. Sydney, E. B. et al. Screening and bioprospecting of anaerobic consortia for biohydrogen and volatile fatty acid production in a vinasse based medium through dark fermentation. *Process. Biochem.* **67**, 1-7 (2018).
42. Luo, J.-F., Lin, W.-T. & Guo, Y. Functional genes based analysis of sulfur-oxidizing bacteria community in sulfide removing bioreactor. *Appl. Microbiol. Biot.* **90**, 769-778 (2011).
43. Willems, A. in *The prokaryotes: Alphaproteobacteria and betaproteobacteria*, E. Rosenberg, E. F. DeLong, S. Lory, E. Stackebrandt, F. Thompson, Eds. (Springer Berlin Heidelberg, Berlin, Heidelberg, 2014), pp. 777-851.
44. Moitinho-Silva, L. et al. Integrated metabolism in sponge–microbe symbiosis revealed by genome-centered metatranscriptomics. *ISME. J.* **11**, 1651-1666 (2017).
45. Márquez, M. et al. *Aquisalibacillus elongatus* gen. Nov., sp. Nov., a moderately halophilic bacterium of the family bacillaceae isolated from a saline lake. *Int. J. Syst. Evol. Microbiol.* **58**, 1922-1926 (2008).
46. de Souza, J. A. M., Carrareto Alves, L., de Mello Varani, A. & de Macedo Lemos, E. G. The family bradyrhizobiaceae. *The Prokaryotes* (Springer: Berlin, Heidelberg, 2014), pp. 135-154. (2014).
47. Bernardet, J.-F. & Nakagawa, Y. An introduction to the family Flavobacteriaceae. *The Prokaryotes: Volume 7: Proteobacteria: Delta, Epsilon Subclass* (Springer New York, NY, 2006), pp. 455-480. (2006).
48. Scheuner, C. et al. Complete genome sequence of *planctomyces brasiliensis* type strain (DSM 5305 T), phylogenomic analysis and reclassification of planctomycetes including the descriptions of *gimesia* gen. Nov., *planctopirus* gen. Nov. And *rubinisphaera* gen. Nov. And emended descriptions of the order planctomycetales and the family planctomycetaceae. *Stand. Genomic. Sci* **9**, 10 (2014).
49. Xuemei, M. J. M. W. L. & Zhang, L. Y. Z. W. 44 The Family Cytophagaceae. *The prokaryotes* (Springer: Berlin, Heidelberg, 2014), pp. 577-593. (2014).